

```

from google import genai

API_KEY = "AQ.Ab8RN6LVp5Rv5iEnIE_nUSF32V8ZnYX7nnOZrXZtd1HrSTvFjA"

client = genai.Client(api_key=API_KEY)

problem = input("Enter the computational problem: ")

prompt = f"""
You are a Python programming expert.

TASK:

Generate a Python program for the following problem:
{problem}

REQUIREMENTS:

1. Use Python.
2. Keep the code simple.
3. Add comments explaining important steps.
4. Use meaningful variable names.
5. The program should accept user input if required.
6. Provide the expected output format.

OUTPUT FORMAT:

1. Problem Explanaton
2. Python Code
3. Sample Input
4. Sample Output
"""

response = client.models.generate_content(
    model="gemini-3.5-fash",
    contents=prompt
)

print("\n===== GENERATED PYTHON PROGRAM =====")

```

```
print(response.text)
```

OUTPUT

```
Python 3.13.7 (tags/v3.13.7:1bceec3, Aug 14 2025, 14:15:11) [MSC v.194
AMD64] on win32
Enter "help" below or click "Help" above for more information.
>> == RESTART: C:/Users/Dell/OneDrive/Desktop/Lab Programs/Generative AI/
Enter the computational problem: Student Problem

===== GENERATED PYTHON PROGRAM =====
### 1. Problem Explanation

**The Student Grade Calculator Problem:**
The goal of this program is to calculate a student's final results bas
r marks in three key subjects: **Math**, **Science**, and **English**.

The program will:
1. Accept the student's name and marks for the three subjects from the
2. Calculate the **total marks** and the **average percentage score**.
3. Assign a letter **grade** based on the following criteria:
   - Average Score  $\geq 90\%$ : **Grade A**
   - Average Score  $\geq 80\%$  and  $< 90\%$ : **Grade B**
   - Average Score  $\geq 70\%$  and  $< 80\%$ : **Grade C**
   - Average Score  $\geq 60\%$  and  $< 70\%$ : **Grade D**
   - Average Score  $< 60\%$ : **Grade F**
4. Display a clean and formatted "Student Report Card".

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### 2. Python Code

'''python
# Program: Student Grade Calculator

# Step 1: Accept student details and subject marks from the user
student_name = input("Enter the student's name: ")

# Convert inputs to float to allow decimal marks
math_score = float(input("Enter marks obtained in Math (out of 100): ")
science_score = float(input("Enter marks obtained in Science (out of 1
english_score = float(input("Enter marks obtained in English (out of 1

xx-cs.py - C:/Users/Dell/OneDrive/Desktop/Lab Programs/Generative AI/xx-cs.py (3.13.7)
File Edit Format Run Options Window Help
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